PROJECT REPORT

Electric Vehicle Market Performance & Insights

Data Analytics with Tableau – Capstone Project

Submitted by: Riya Sharma

Medicaps University

Team ID: NA

November 2026

Dashboard: https://public.tableau.com/app/profile/riya.sharma8766/viz/ev_riya_final/EVTechnologyDashboard

Story: https://public.tableau.com/app/profile/riya.sharma8766/viz/ev_riya_final/Story1

GitHub: [Paste GitHub repository link]

Demo video: [Paste Google Drive demo video link]

1. INTRODUCTION

1.1 Project Overview

Electric vehicle adoption depends not only on sales but on technology – battery size, energy efficiency, charging time, motor power – and on ownership costs such as insurance. Stakeholders need to see these together with regional sales.

In this project I used the cleaned EV dataset (199,999 records, 37 attributes, model years 2018–2026) and built an EV Technology Dashboard in Tableau with a KPI strip, regional and technology charts, two extra analysis sheets and a 3-point story, published on Tableau Public.

1.2 Purpose

The dashboard answers the questions of three users:

- **Emily Johnson (EV Market Strategy Director):** needs to know which segments perform best, how battery size and efficiency differ by category, and where sales are growing, to plan the product portfolio.
- **David Chen (Automotive BI Manager):** compares regions, battery technologies, drivetrains and ownership costs so he can recommend markets and products to his clients.
- **Dr. Sarah Williams (Sustainable Mobility Research Analyst):** researches battery technology, energy efficiency and charging, and how they differ across regions and vehicle types, to guide clean-transport policy.

2. IDEATION PHASE

2.1 Problem Statement

Customer Problem Statements

Customer Problem Statement template (Miro)

PS-1

I am a Market Strategy Director (Emily Johnson).
I'm trying to see how regions and battery types perform together.
But sales and technical specs sit in different columns.
Because there is no view combining region, battery and category.
Which makes me feel stressed and unsure before planning meetings.

PS-2

I am an Automotive BI Manager (David Chen).
I'm trying to advise clients on drivetrains, battery types and ownership cost.
But I must compare price, insurance and motor power by hand.
Because there is no interactive tool that links them.
Which makes me feel slow and less confident.

PS-3

I am a Sustainable Mobility Researcher (Dr. Sarah Williams).
I'm trying to compare battery size, energy efficiency and charging time.
But these specs differ widely between vehicle types.
Because existing reports do not compare categories visually.
Which makes me feel frustrated and unable to give quick policy advice.

2.2 Empathy Map Canvas

Empathy Map – EV Technology Dashboard Users

Users: Emily Johnson (Strategy Director) · David Chen (BI Manager) · Dr. Sarah Williams (Research Analyst)

SAYS
- "Which segments need the biggest batteries?"
- "Is demand stronger in some regions?"
- "Which vehicles use the most energy per km?"
- "How do price and insurance compare?"

THINKS
- Technical specs are spread over many columns
- Do battery types differ by region?
- Does chemistry change charging time?
- Are heavy EVs efficient enough?

DOES
- Compares spec sheets by hand
- Makes separate charts per region
- Reads battery and charging studies
- Estimates ownership costs manually

FEELS
- Overloaded with technical data
- Unsure which region to focus on
- Rushed before client meetings
- Confident when specs are compared visually

PAINS
- No combined view of specs, regions and sales
- Slow manual comparison
- Hard to see region × battery patterns
- Cost of ownership not visible

GAINS
- One technology dashboard with KPIs and 6 charts
- Filters for region, category and model year
- Battery, efficiency and charging shown visually
- Story and public link for stakeholders

2.3 Brainstorming

Ideas were ranked by feasibility and importance. A regional sales dashboard with KPIs, technology views (battery, efficiency, charging) and cost/performance sheets with a story were selected; a mobile app was not selected.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Customer Journey Map – EV Technology Dashboard User

Persona: Emily Johnson (also representing David Chen and Dr. Sarah Williams)

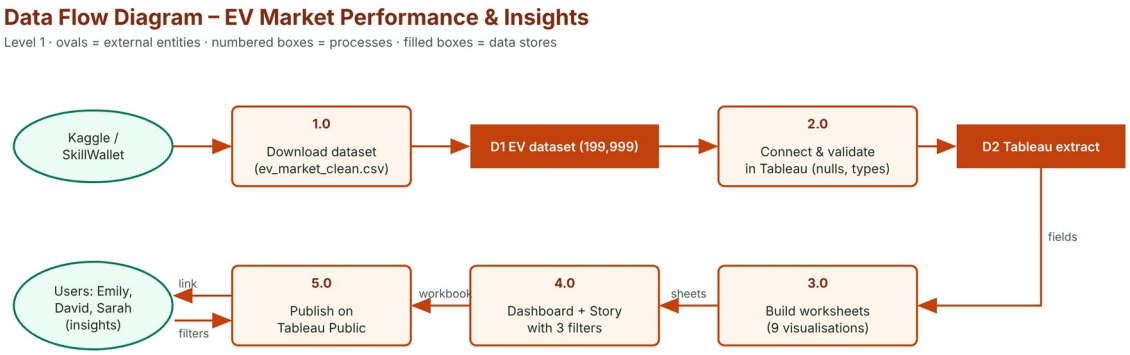
	1. Question	2. Explore	3. Act
Actions	Gets a question on regions or vehicle technology	Uses region, category and model-year filters	Shares the story and recommends actions
Touchpoints	Meeting / email, raw dataset	KPIs, treemap, trend, heatmap, battery and efficiency charts	Tableau Story, Tableau Public link
Thoughts	"How do I compare all these specs?"	"NMC/LFP lead everywhere; buses need huge batteries."	"Now I can advise with evidence."
Feelings	😟 Stressed	🙂 Interested	😊 Confident
Ownership	Business user	Analyst (Riya Sharma)	Business user
Opportunities	Keep data in one place	Add more filters and tooltips	Add forecasting in future

3.2 Solution Requirement

FR	Requirement	Description
FR-1	Data access	Connect ev_market_clean.csv to Tableau as an extract
FR-2	Data processing	Cleaned data; calculated fields
FR-3	Analysis	SUM and AVG by region, category, battery type, drivetrain

FR-4	Outputs	9 worksheets, dashboard, story
FR-5	Interaction	Region, category and model-year filters
FR-6	Reporting	Tableau Public link

3.3 Data Flow Diagram

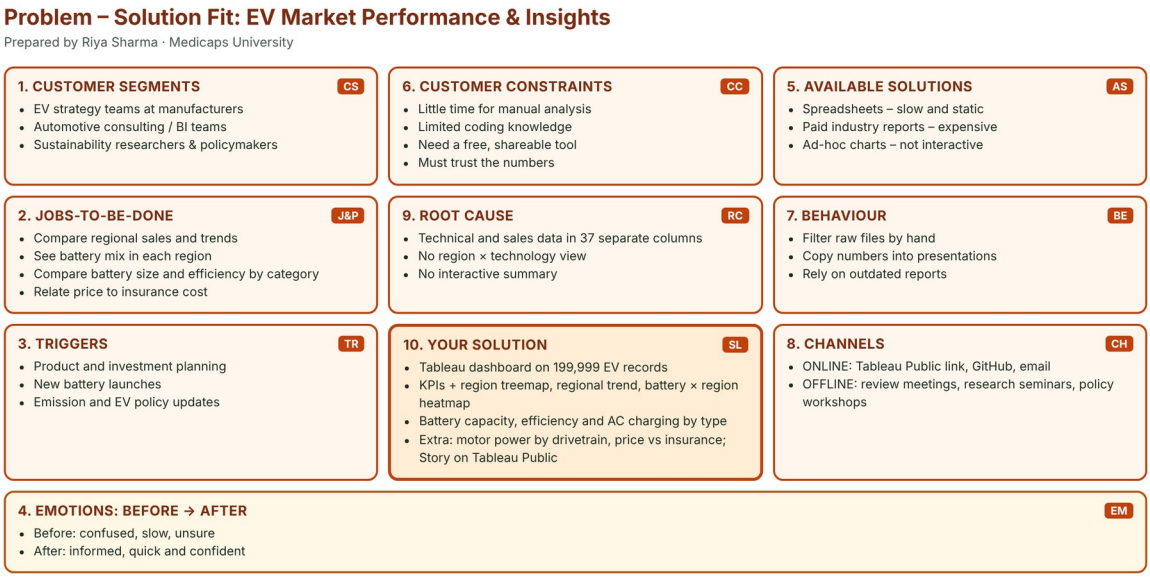


3.4 Technology Stack

Tableau Desktop / Tableau Public for building and publishing; cleaned CSV + Tableau extract for data; 7 Tableau calculated fields for KPIs; GitHub for files and documentation; Google Drive for the demo video.

4. PROJECT DESIGN

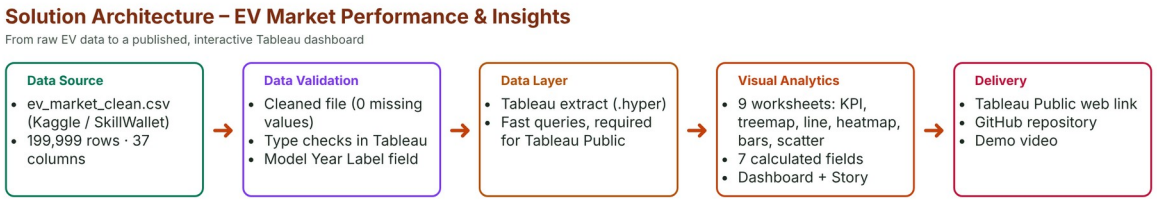
4.1 Problem Solution Fit



4.2 Proposed Solution

Cleaned data → extract → calculated fields → 9 worksheets → dashboard with 3 shared filters → story → Tableau Public → testing and documentation.

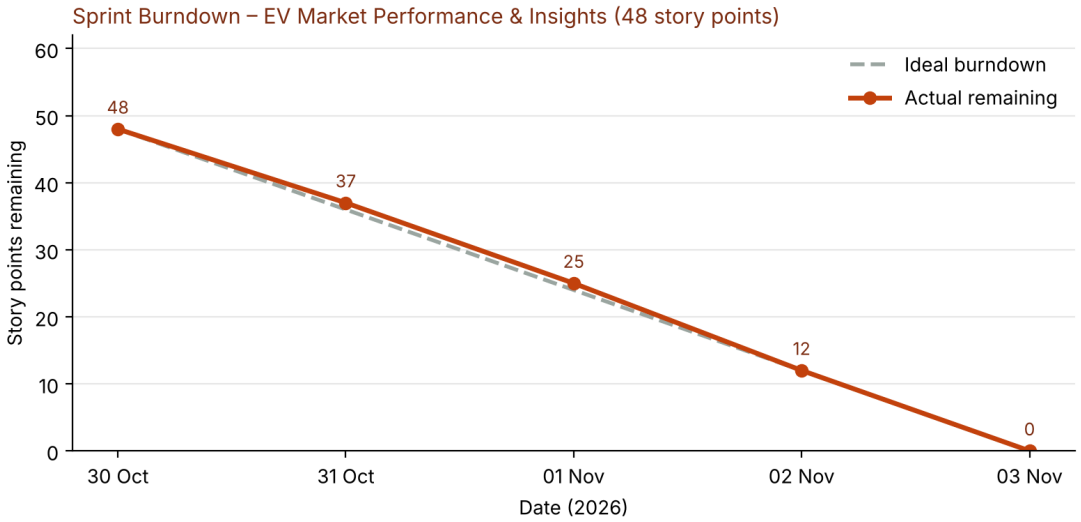
4.3 Solution Architecture



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

19 user stories (48 story points) were completed in 4 one-day sprints between 30 October and 3 November 2026; average velocity 12 points per sprint.



6. FUNCTIONAL AND PERFORMANCE TESTING

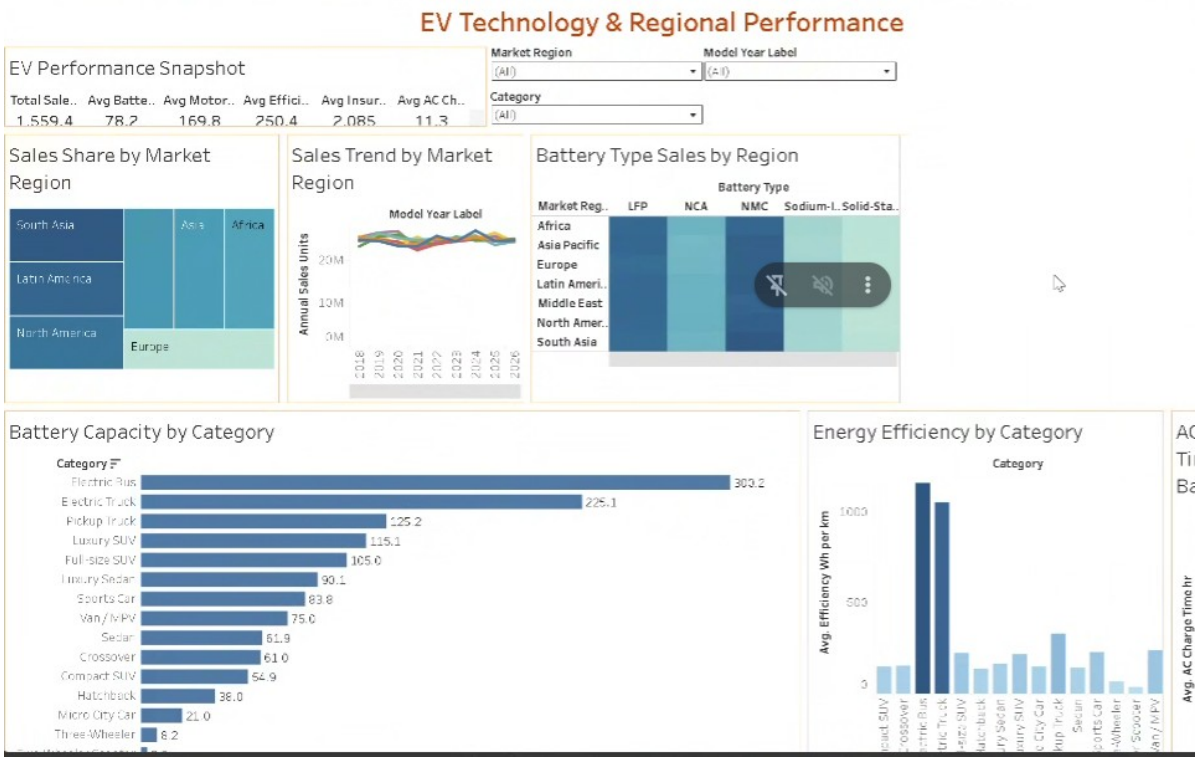
6.1 Performance Testing

Parameter	Values
Data Rendered	199,999 rows × 37 columns + 7 calculated fields
Data Quality	0 duplicates, 0 missing values
Filters	Market Region, Category, Model Year
Calculations	Total Sales (M Units); Avg Battery (kWh); Avg Motor Power (kW); Avg Efficiency (Wh/km); Avg Insurance (USD/yr); Avg AC Charge (hrs); Model Year Label
Dashboard	KPIs + 6 charts; 9 worksheets in total

Story	3 story points

7. RESULTS

7.1 Output Screenshots



EV Technology Dashboard

1. EV Performance Snapshot

Question: What are the headline technical and sales figures of the EV market? (Scenario 1, 2, 3)



Insight: Total annual sales are 1,559.4 M units; an average EV has a 78.2 kWh battery and a 169.8 kW motor, uses 250.4 Wh per km, costs 2,085 USD a year to insure and needs 11.3 hours to charge on AC.

2. Sales Share by Market Region

Question: How is EV sales volume shared between market regions? (Scenario 2)



Insight: Every region holds about 14% of sales: South Asia (225.2 M units) and Latin America (224.7 M) are slightly ahead, Europe (218.8 M) is slightly behind – demand is balanced worldwide.

3. Sales Trend by Market Region

Question: How have sales in each region changed over model years 2018–2026? (Scenario 1, 2)



Insight: All seven regional lines move closely together; total sales stay between 166.7 M and 178.0 M units per model year, with peaks for 2019 and 2024 models – growth is steady in every region.

4. Battery Type Sales by Region

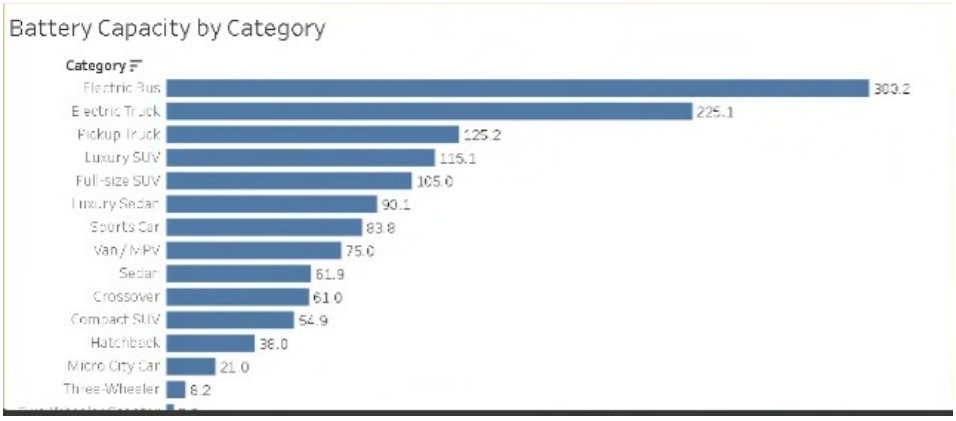
Question: Which battery technologies sell most in each region? (Scenario 2, 3)



Insight: NMC and LFP are the darkest cells in every region (about 70–79 M units each), with NMC strongest in South Asia (78.5 M). Sodium-Ion and Solid-State are the lightest (15–23 M) – new chemistries are still small everywhere.

5. Battery Capacity by Category

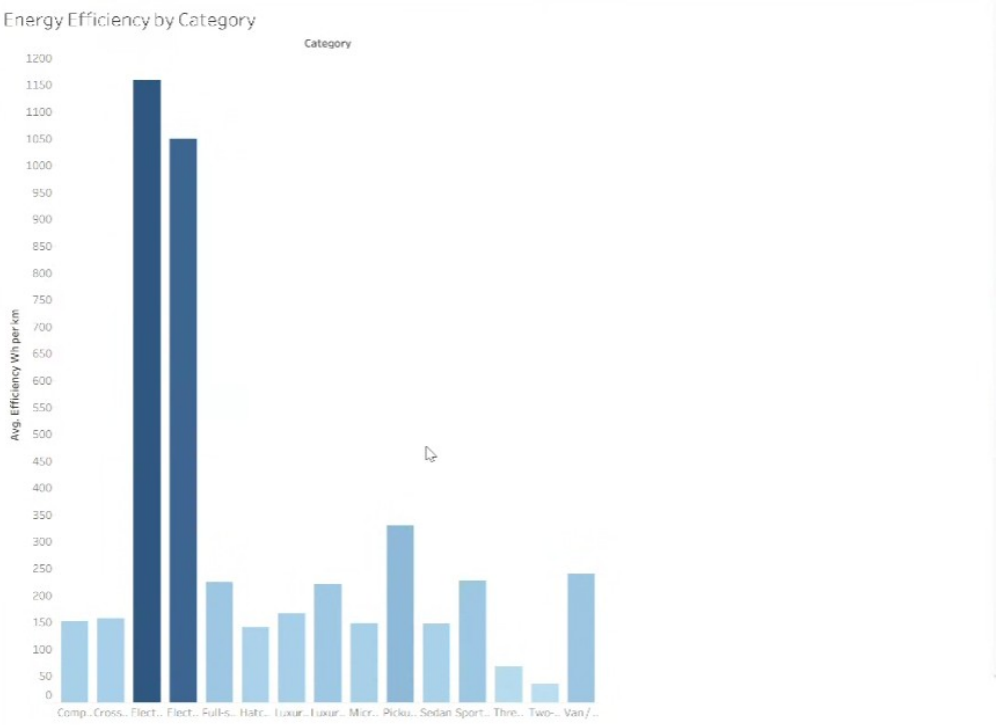
Question: How large are the batteries in each vehicle category? (Scenario 1, 3)



Insight: Electric Buses carry the biggest batteries (300.2 kWh), followed by Electric Trucks (225.1 kWh) and Pickup Trucks (125.2 kWh). Two-Wheeler Scooters need only 3.3 kWh.

6. Energy Efficiency by Category

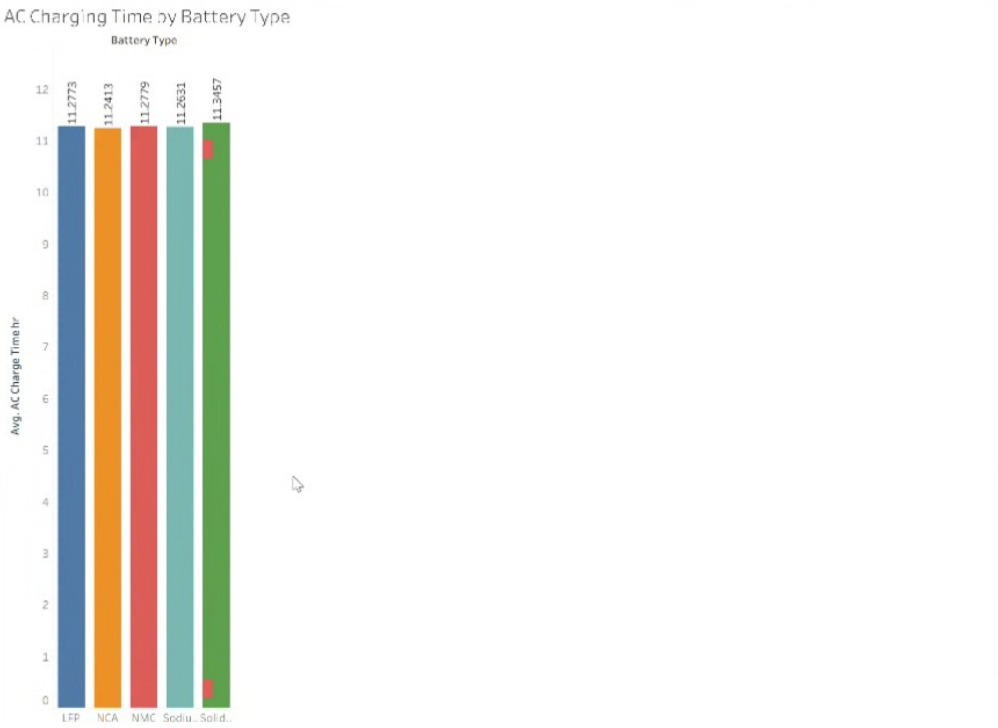
Question: Which vehicle categories use the most energy per kilometre? (Scenario 3)



Insight: Electric Buses (1,159 Wh/km) and Electric Trucks (1,049 Wh/km) use far more energy per km than passenger cars (≈140–240 Wh/km), while Two-Wheelers use only 35 Wh/km.

7. AC Charging Time by Battery Type

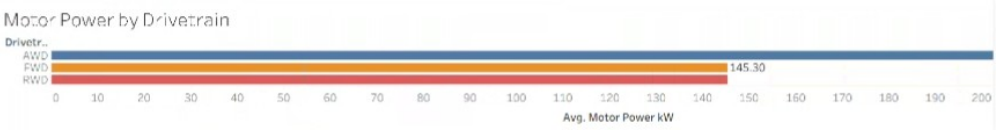
Question: Does battery chemistry change AC charging time? (Scenario 3)



Insight: Average AC charging time is almost the same for every chemistry – about 11.2–11.3 hours (Solid-State 11.35 h, NCA 11.24 h). Charging time depends on battery size, not chemistry.

8. Motor Power by Drivetrain

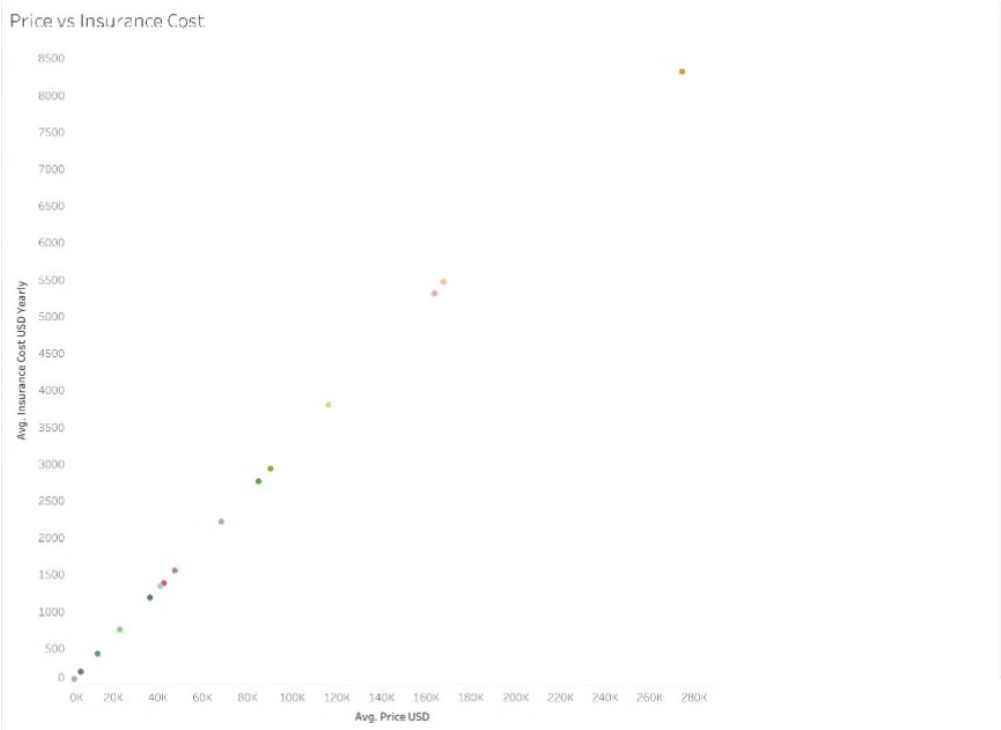
Question: How does motor power differ between AWD, FWD and RWD? (Scenario 1, 2)



Insight: AWD vehicles have the most powerful motors (223.8 kW on average), about 54% more than FWD and RWD vehicles (145.3 kW each).

9. Price vs Insurance Cost

Question: How is vehicle price related to yearly insurance cost? (Scenario 2)



Insight: Insurance rises almost in a straight line with price (correlation 0.92): Two-Wheelers cost about 2,555 USD and 85 USD/yr to insure, while Electric Buses cost about 274,543 USD and 8,316 USD/yr.

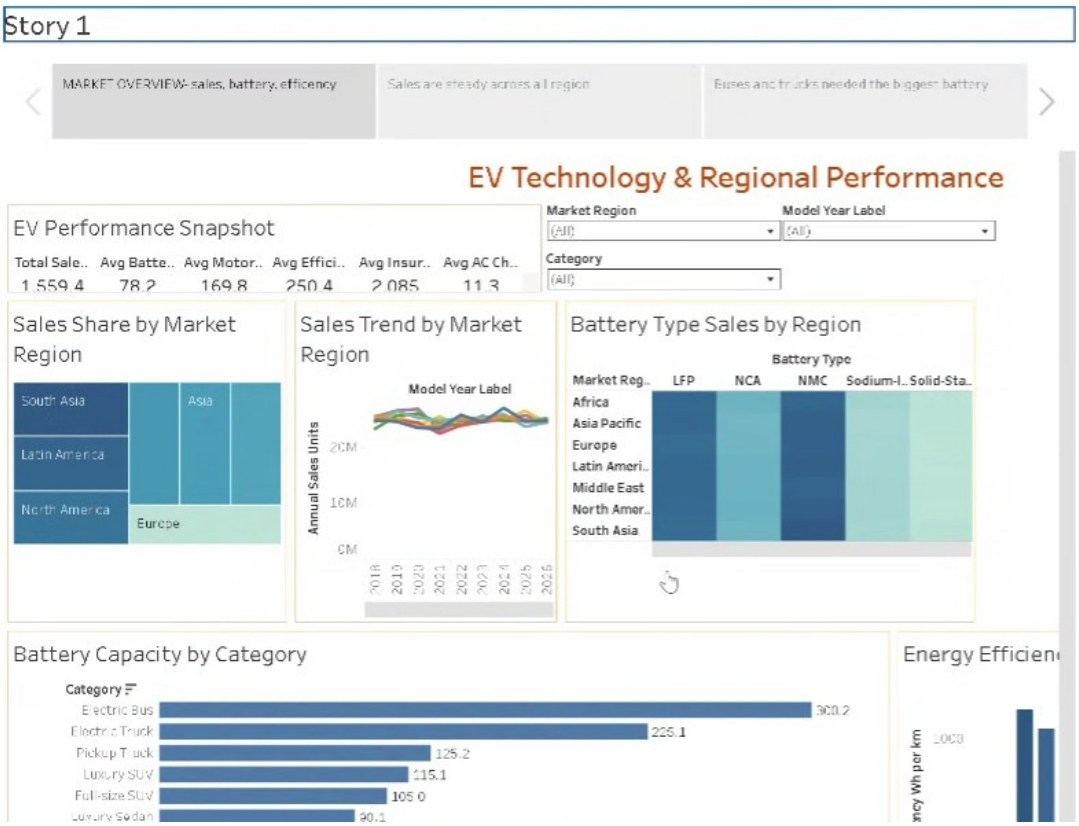


Tableau Story

8. ADVANTAGES & DISADVANTAGES

Advantages

- Combines sales, technology and cost in one dashboard.
- Three shared filters for focused analysis.
- Free and shareable through Tableau Public.
- Story makes findings easy to present.

Disadvantages

- Static data file, no live refresh.
- Some dashboard charts are small on laptop screens.
- No forecasting yet.

9. CONCLUSION

The EV Technology Dashboard shows that sales are evenly spread across the seven regions and steady over time, NMC and LFP are the leading batteries in every region, buses and trucks need by far the largest batteries and the most energy per km, AC charging takes about 11.3 hours whatever the chemistry, AWD vehicles have about 54% more motor power, and insurance cost rises almost linearly with price. These insights support Emily's portfolio planning, David's client advice and Sarah's research.

10. FUTURE SCOPE

- Drill-down from region to country.
- Forecast sales by region.
- Live data with scheduled refresh.
- Total cost of ownership analysis.

11. APPENDIX

Item	Link
Dataset	https://www.kaggle.com/datasets/tharishreddy22/electric-vehicle-market-classification-dataset
Dashboard	https://public.tableau.com/app/profile/riya.sharma8766/viz/ev_riya_final/EVTechnologyDashboard
Story	https://public.tableau.com/app/profile/riya.sharma8766/viz/ev_riya_final/Story1
GitHub	[Paste GitHub repository link]
Demo video	[Paste Google Drive demo video link]

Data preparation:

- Used the cleaned EV dataset ev_market_clean.csv: 199,999 rows × 37 columns (prepared from the Kaggle / SkillWallet ev_market_train.csv file).
- Cleaning already applied to this file: 1 duplicate row removed and about 19,000 missing values filled with category medians, so it has 0 missing values.
- Checked data types in Tableau's Data Source tab; created a Model Year Label field so model years work as categories.
- Created a Tableau extract (.hyper) so the dashboard is fast and can be published to Tableau Public.